

**A Multifaceted Exploration of Spatial
Openness in Rental Housing: Big Data
Analysis Across Tokyo's 23 Wards**

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Abstract

Understanding spatial openness in residential housing is important for enhancing living quality and guiding architectural design. This study proposes a multifaceted evaluation method of spatial openness in rental housing by leveraging large-scale data from Tokyo's 23 wards. Visibility Graph Analysis (VGA) is applied to floor plan images to assess two-dimensional spatial connectivity. In addition, semantic segmentation is performed on interior photos to capture three-dimensional openness features such as window ratios and ceiling visibility. The study investigates how openness characteristics vary by geographic location and construction year, identifying spatial and temporal patterns. Finally, it explores the relationship between these openness features and subjective housing preferences, providing insights into how spatial openness influences perceived residential quality.

Acknowledgements

This is a good place to acknowledge those who helped you along the way.

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Chapter 1

Introduction

1.1 Research Background and Motivation

Understanding spatial openness in residential housing is fundamental to enhancing living quality and guiding architectural design practices. In urban environments, particularly in densely populated cities like Tokyo, the perception of spatial openness significantly impacts residents' psychological well-being, comfort, and overall satisfaction with their living spaces. Traditional architectural evaluation methods have long relied on basic quantitative metrics such as floor area, room count, and ceiling height to assess spatial quality. However, these conventional indicators fail to capture the nuanced spatial qualities that contribute to the subjective experience of openness in residential environments.

The concept of spatial openness extends beyond mere physical dimensions to encompass the complex interplay between visual connectivity, visibility of interior elements, and so on. In the context of rental housing markets, where prospective tenants make decisions based on limited visual information from online platforms, understanding how spatial openness translates through digital representations becomes increasingly important. The proliferation of online rental platforms has created unprecedented opportunities to analyze large-scale housing data, including interior imagery and floor plans, which can provide insights into

spatial characteristics that were previously difficult to quantify systematically.

Tokyo’s 23 special wards represent a unique data playground for studying spatial openness due to the city’s distinctive urban morphology, diverse architectural styles spanning multiple decades, and the comprehensive digitization of rental housing information, which has always been one of the most populated cities and rental market in the world.

1.2 Literature Review and Research Gaps

1.2.1 Spatial Openness in Architectural Research

The study of spatial openness has been approached from various perspectives within architectural and environmental psychology research. Traditional spatial analysis methods have focused on geometric properties and numerical metrics of building attributes. Space syntax theory, developed by Hillier and Hanson, introduced systematic approaches to understanding spatial relationships through graph-based analysis of architectural layouts [1]. Visibility Graph Analysis (VGA), as an extension of space syntax principles, provides a quantitative framework for measuring visual connectivity and spatial integration within architectural spaces.

Previous research has demonstrated that spatial openness influences occupant behavior, movement patterns, and psychological responses to built environments. Studies in environmental psychology have shown that perceived openness correlates with reduced stress levels, improved cognitive performance, and enhanced overall well-being. However, most existing research has been limited to small-scale case studies or controlled experimental settings, lacking the scale and diversity necessary to understand spatial openness patterns across entire urban regions.

1.2.2 Computer Vision and Spatial Analysis

Recent advances in computer vision and deep learning have opened new possibilities for automated spatial analysis. Semantic segmentation techniques enable the extraction of architectural elements from images, allowing for systematic analysis of interior compositions. Similarly, these deep-learning-based models have been successfully applied to recognize and classify various architectural features in floor plans, including walls, windows, floors, and ceilings. These technological developments create opportunities to process large-scale image datasets that would be impractical to analyze manually.

The integration of 2D image analysis with 3D spatial analysis recently draws more attentions in architectural research. While 2D analysis can capture visual qualities and material compositions, 3D analysis provides insights into spatial relationships and configurational properties. The combination of these approaches offers a more comprehensive understanding of spatial openness than either method alone.

1.2.3 Big Data Opportunities and Research Gaps

The emergence of big data analytics in housing research has revealed new patterns in residential preferences, market dynamics, and architectural trends. Online rental platforms generate vast amounts of data, including property specifications, pricing information, and visual documentation through interior images and floor plans. This data richness enables researchers to investigate questions about spatial quality and market valuation at unprecedented scales, creating opportunities that were previously impossible due to data limitations.

However, the potential for using big data to understand subjective spatial experiences remains largely unexplored, particularly in the context of rental housing markets that traditionally rely on rule-based tabular data without in-depth understanding of spatial qualities. While studies using conventional tabular data can achieve large scale, spatial openness research involving image analysis or human-operated software assessments has been limited to small samples, preventing generalization to broader urban contexts. Additionally, the tools

and methodologies for systematically processing and analyzing large-scale visual and spatial data in housing research are still developing, creating both opportunities and challenges for comprehensive spatial analysis.

1.3 Primary Research Objectives

This research aims to address the identified gaps through the following primary objectives:

1. **Develop a Comprehensive Methodology:** Create a multifaceted evaluation method for spatial openness that combines automated extraction of visual elements from interior images with floorplan semantic segmentation for VGA simulations, enabling large-scale analysis previously limited by manual assessment constraints.
2. **Large-Scale Analysis:** Conduct systematic analysis of spatial openness using large-scale rental housing data from Tokyo’s 23 wards.
3. **Temporal and Geographic Investigation:** Examine how spatial openness varies across construction periods and geographic regions within Tokyo’s 23 wards.
4. **Feature Dependency Analysis:** Investigate relationships between 2D visual elements and 3D spatial connectivity measures to identify key interactions contributing to spatial openness perception.
5. **Market Correlation Analysis:** Explore the relationships between spatial openness features, impression evaluations and rental costs.

1.4 Structure of the Thesis

This thesis is organized into six main chapters, each addressing specific aspects of spatial openness analysis:

Chapter 1: Introduction provides the research background, literature review, objectives, and methodology overview.

Chapter 2: Methodology presents detailed descriptions of the Lifull dataset, openness quantification methods including interior image-based 3D perception analysis and floor plan-based 2D VGA analysis, and statistical analysis frameworks.

Chapter 3: Results presents the primary findings, including temporal analysis of openness characteristics across construction decades and geographic distribution analysis across Tokyo's 23 wards.

Chapter 4: Comprehensive Openness Analysis provides in-depth analysis including regression analysis against rent costs and detailed analysis of openness features, with image examples and comparison between 2D and 3D openness measures.

Chapter 5: Discussion interprets the results, discusses implications for architectural design and urban planning, addresses limitations, and suggests future research directions.

Chapter 6: Conclusion summarizes the key findings, contributions, and practical implications of the research.

The thesis concludes with comprehensive appendices containing detailed technical specifications, additional statistical analyses, and supplementary materials that support the main findings.

This research provides a framework for analyzing spatial openness in residential environments, with potential applications in architectural design and urban planning.

Chapter 2

Methodology

This chapter presents the data filtering process, dataset characteristics, and methodology for quantifying spatial openness from interior images and floor plan data. The openness features extractions include the windows ratio, ceilings ratio, floors ratio, and walls ratio from interior images, together with tabular based attributes of properties. VGAs relies on the floorplan data. These features are used in subsequent temporal, spatial, and correlation analyses.

2.1 Overview of the LIFULL Real Estate Dataset

Our dataset comprises 4004 rental properties from Tokyo’s 23 wards (1960-present), filtered for image availability and completeness. The LIFULL dataset represents one of Japan’s largest real estate databases, providing comprehensive information about rental properties including interior images, floor plan data, and property specifications.

2.1.1 Dataset Characteristics

The dataset encompasses multiple categories of spatial openness attributes through interior images, floor plan data, and property specifications. Key characteristics include:

- **Temporal Coverage:** Properties spanning from 1960 to present, allowing for longitudinal analysis of spatial design trends
- **Geographic Scope:** Complete coverage of Tokyo’s 23 special wards, providing urban-scale spatial analysis capabilities
- **Data Completeness:** Properties filtered for availability of both interior images and floor plan data

2.1.2 Data Structure and Availability

The dataset provides multi-modal information for each property:

- **Interior Images:** High-resolution photographs of living spaces, where we focus on the living room, as shown in Figure 2.1.
- **Floor Plans:** Architectural drawings showing spatial layout and room configurations, as shown in Figure 2.2.
- **Property Specifications:** Detailed attributes including room size, building age, rent prices, and other basic tabular specs, as exemplified in Table 2.1. addresses.

Table 2.1: Sample of LIFULL dataset structure showing key property attributes

Property ID	Prefecture	City/Ward	...	Building Area	Rent Price
00026ec12fcea938...	Tokyo	Shibuya	...	51.0	88000
00040e134334f56c...	...	...	...	...	...

2.2 Definition of Openness

Spatial openness is typically expressed through the sense of visual connectivity. In residential rental markets, the 3D visual elements displayed in property photographs serve as the primary carriers of spatial impression and openness perception for potential tenants. The



Figure 2.1: Example of interior image from the LIFULL dataset.

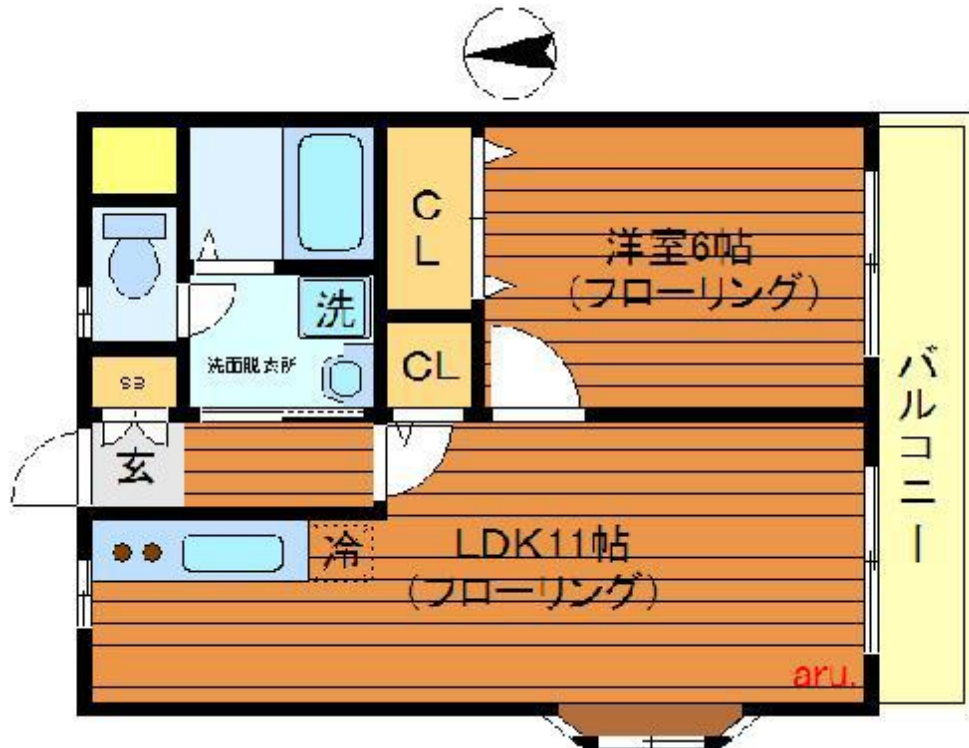


Figure 2.2: Example of floorplan image from the LIFULL dataset.

concept of openness in residential spaces encompasses several dimensions, with particular emphasis on the following:

- **Visual Connectivity:** The degree to which different areas of a space are visually connected as perceived through interior photographs, which numerically quantified by the VGA analysis.
- **Architectural Elements:** The appearance of windows, ceilings, floors, and walls in photos of the living rooms in the LIFULL dataset.

In the context of rental property advertising, these visual elements become the primary means through which spatial qualities are communicated to potential tenants, making their accurate quantification essential for understanding market dynamics.

2.3 Data Filtering and Cleaning

In order to effectively extract the required openness features, the data filtering and cleaning pipeline ensures data quality and consistency across the dataset. Multiple stages of filtering and validation are applied to maintain analytical reliability. The following sections detail the filtering process for interior images, floor plans, and property specifications, which are also illustrated as shown in Figure 2.3. The details of the methodologies of how to extract the openness features are demonstrated later after stating the filtering process.

2.3.1 Multi-Stage Data Filtering Process

Starting from a raw dataset of over 5 million properties, a systematic multi-stage filtering process is applied to ensure data quality and analytical reliability. The filtering pipeline reduces the dataset to approximately 4,000 final samples suitable for comprehensive openness analysis.

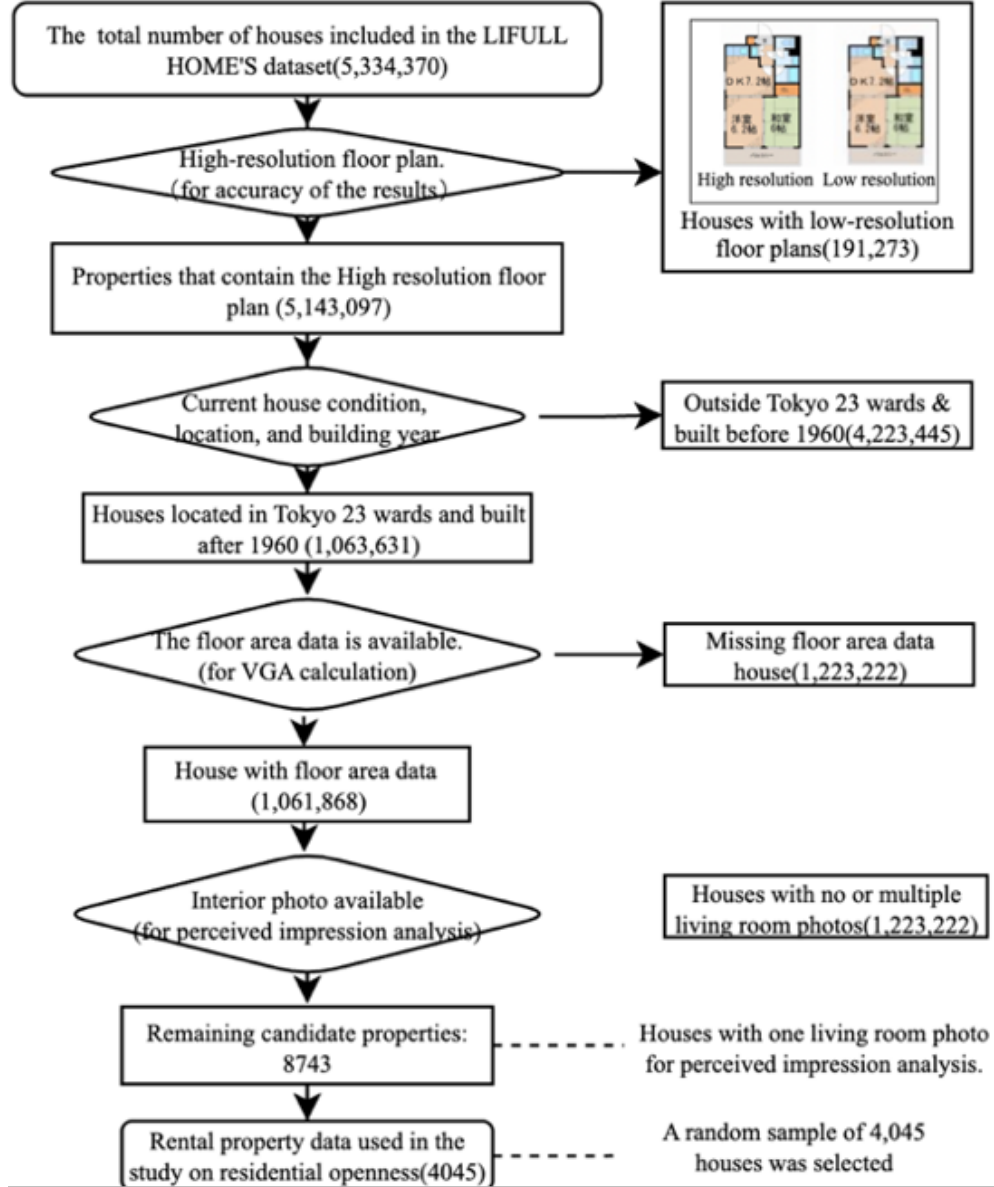


Figure 2.3: Data filtering and preprocessing pipeline showing the systematic approach to ensuring data quality and consistency across interior images, floor plans, and property specifications. The flowchart illustrates the multi-stage filtering process from raw data collection to the final integrated dataset.

Initial Data Selection

The first stage focuses on selecting properties with high-quality visual data:

- **High-Resolution Floor Plans:** Selection of properties with clear, high-resolution floor plan data only, excluding low-resolution variants to ensure adequate spatial detail for VGA analysis and improved semantic segmentation performance
- **Clear Interior Images:** Properties filtered to include only those with clear living room photographs available, ensuring consistent performance when extracting architectural element ratios through semantic segmentation
- **Geographic Constraint:** Properties limited to Tokyo's 23 wards with accurate location data

Property Specification Requirements

Properties must contain complete tabular specifications including:

- **Location Information:** Accurate geographic coordinates and ward designation within Tokyo's 23 wards
- **Construction Details:** Building construction year, building area, and property type specifications
- **Room Configuration:** Properties must contain room open space area information and at least 1DK room configuration for spatial analysis
- **Essential Attributes:** Complete set of required building-related information for comprehensive analysis

Quality Validation and Final Selection

The final filtering stage ensures analytical validity:

- **VGA Results Validation:** Properties filtered based on legitimate availability of VGA analysis results from floor plan processing
- **Architectural Element Detection:** Verification that semantic segmentation successfully identifies at least one of the key spatial elements (windows, ceilings, floors, or walls) within interior images
- **Multi-Modal Integration:** Final verification that properties contain all required data modalities (high-resolution floor plans, clear interior images, and complete specifications) for comprehensive openness analysis

This systematic filtering process ensures that the final dataset of approximately 4,000 properties maintains high data quality standards while providing sufficient sample size for robust statistical analysis. The following sections detail the extraction process of the openness features that have been already mentioned in the data filtering and cleaning workflow.

2.4 Openness Features Extractions

2.4.1 Visibility Graph Analysis (VGA) Methodology

A key metric for measuring spatial openness is the VGA (Visibility Graph Analysis) value from floor plans, which reflects spatial connectivity. The VGA methodology provides a quantitative approach to assess spatial relationships and visual connections within architectural layouts, complementing the impression-based analysis derived from interior photographs.

Traditionally, VGA analysis requires the transformation of optical floor plans into vectorized data formats that can be processed using specialized third-party software such as depthmapX [2]. This conversion process involves:

- **Digitization:** Converting raster floor plan images into vector-based geometric representations

- **Spatial Calibration:** Ensuring accurate scale and dimensional relationships in the vectorized format
- **Boundary Definition:** Precise delineation of walls, openings, and spatial boundaries for visibility calculations
- **Software Processing:** Utilization of specialized tools like depthmapX for comprehensive spatial analysis

However, this traditional approach presents significant scalability challenges when applied to large-scale datasets. Each floor plan requires individual attention for accurate conversion, and spatial, calibration, showing a bottleneck that limits its applicability to larger data samples due to the extensive human operations involved.

Mathematical Definition of VGA

Corresponding to the step 4 in the above section, the VGA value quantifies the degree of spatial integration and visual connectivity within a given floor plan layout. For a discretized spatial grid with N points, the visibility graph analysis computes connectivity measures based on line-of-sight relationships between grid points.

Mathematically, the VGA value at each point is defined as:

$$S(i) = \sum_{j=1}^N V_{ij} \quad (2.1)$$

where:

- $S(i)$ represents the visibility score at grid point i
- V_{ij} is the visibility indicator function defined as:

$$V_{ij} = \begin{cases} 1 & \text{if point } j \text{ is visible from point } i \\ 0 & \text{otherwise} \end{cases} \quad (2.2)$$

- N is the total number of grid points in the open space

The visibility determination between two points follows the isovist principle, where point j is considered visible from point i if the straight line connecting them does not intersect any wall boundaries in the floor plan.

This computation generates a VGA heatmap $\mathbf{S} = \{S(1), S(2), \dots, S(N)\}$ representing spatial connectivity throughout the floor plan. From this heatmap, summary statistics including mean (μ_S), standard deviation (σ_S), and percentiles are extracted as quantitative features characterizing spatial openness.

VGA Computation Process

To address traditional VGA limitations, we developed a computational workflow using optical floor plan images directly:

1. **Semantic Segmentation:** DeepLab-V3+ model [3] fine-tuned to classify walls, bedrooms, and living rooms
2. **Boundary Mask Creation:** Generate masks distinguishing walls from open spaces
3. **Physical Grid Generation:** Create uniform grid overlay on open space areas
4. **Isovist-based Visibility Calculation:** Calculate visibility for each grid point using the isovist method
5. **Statistical Extraction:** Extract statistical measures from the resulting heatmap to quantify spatial connectivity

This workflow is demonstrated in Figure 2.4.

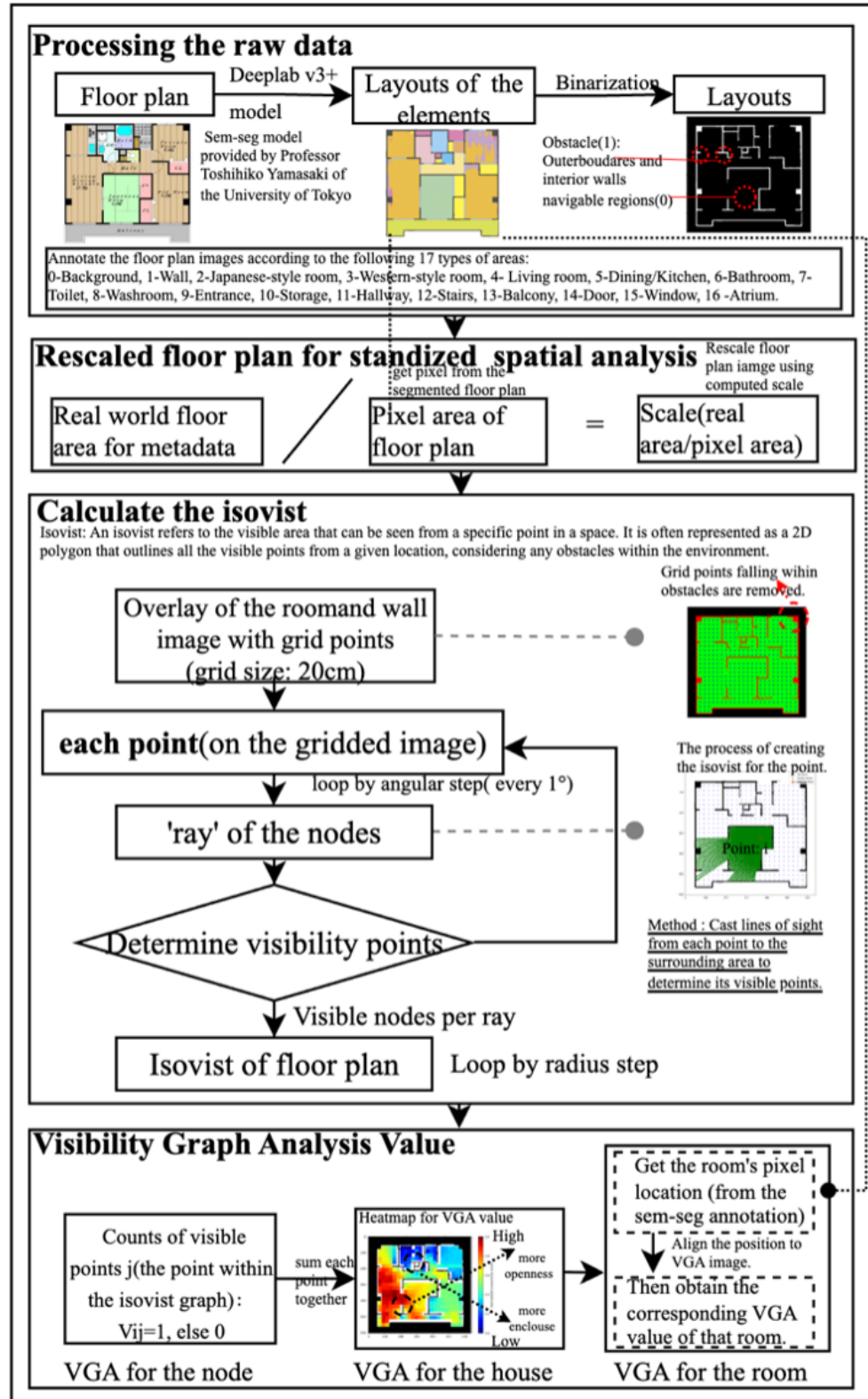


Figure 2.4: VGA computational workflow demonstrating the step-by-step process from semantic segmentation of floor plans to statistical analysis of spatial connectivity.

2.4.2 Architectural Element Ratios

Beyond VGA analysis, spatial openness is quantified through the analysis of key architectural elements extracted from interior images. In our studies, such ratios are calculated from the semantic segmentation results of living room images.

- **Windows Ratio:** Proportion of window area relative to total visible area, indicating natural light access and external connectivity
- **Ceilings Ratio:** Ceiling visibility and height perception contributing to vertical spatial openness
- **Floors Ratio:** Floor area visibility affecting spatial grounding and continuity
- **Walls Ratio:** Wall presence and configuration influencing spatial division and flow

These ratios are derived through semantic segmentation techniques applied to interior imagery, providing quantitative measures of spatial composition. However, such extraction may have intrinsic uncertainty due to the photo angles and standing positions collected from the website, but there is hardly any way to eliminate such issues when requesting to use large-scale data from Lifull which is not designed specifically for this study. The methodological details are demonstrated in the following sections.

Segmentation Model

The semantic segmentation employs a Mask2Former pre-trained model [4] optimized for architectural element recognition.

- **Model Selection:** Utilization of state-of-the-art segmentation architectures finetuned on interior image datasets
- **Element Categories:** Classification of pixels into windows, ceilings, floors, walls, and other architectural components

Pixel-wise Classification

The segmentation process generates pixel-wise masks for each architectural element:

1. **Image Preprocessing:** Standardization of input images for consistent processing, including resizing and batching.
2. **Segmentation Execution:** Application of the trained model to generate semantic masks
3. **Post-processing:** recover the original shape and resolutions of the semantic masks to properly calculate the ratios
4. **Refinement Process:** Iterative results by ensuring at least one of the four architectural elements (windows, ceilings, floors, walls) appears in the image.

The above workflow is demonstrated in Figure 2.5.

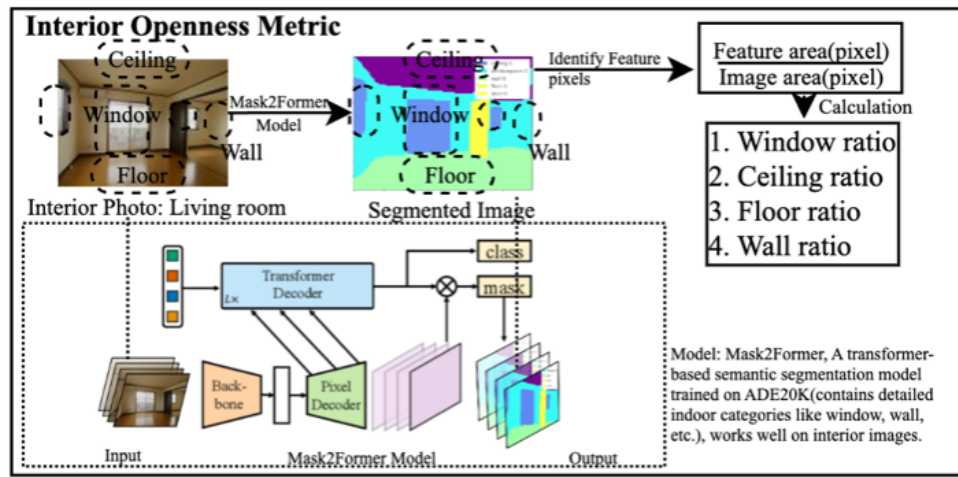


Figure 2.5: Semantic segmentation workflow for architectural element extraction from interior images, showing the process from input image through pixel-wise classification to architectural element ratio calculation.

Ratio Calculation

For each architectural element, the ratio is computed as:

$$\text{Element Ratio} = \frac{\text{Element Pixel Count}}{\text{Total Image Pixels}} \quad (2.3)$$

This provides a normalized measure of each element's presence within the visual field.

Having established the comprehensive data preparation pipeline and feature extraction methodologies, we now transition next chapter for the analytical framework designed to examine the relationship between extracted openness features and human perceptions of spatial openness. This analysis encompasses both individual feature impacts and their collective influence on openness evaluations.

Chapter 3

Quantifying the Openness of Residential Spaces

Temporary placeholder citation to satisfy BibTeX: [5]

3.1 Floor Plan Processing and Visibility Graph Analysis (VGA)

3.2 Openness Feature Extraction from Interior Images

- Estimating Window Proportions
- Segmenting Ceilings, Walls, Doors, etc.

3.3 Matching Strategy for Floor Plans and Interior Images

3.4 Visualization Examples of Openness Features

Chapter 4

Influence of Age and Geographic Factors

- 4.1 Impact of Building Age on Openness-Impression Relationship
- 4.2 Comparative Analysis Across Geographic Areas
- 4.3 Spatial Visualization and Cluster Analysis

Chapter 5

Correlation Analysis Between Subjective Impressions and Openness

5.1 Impression Score Outputs and Interpretation

5.2 Correlation Between Openness Features and Subjective Impressions

5.3 Case Studies and Visual Comparisons

Chapter 6

Conclusion and Future Work

6.1 Summary of Findings and Contributions

6.2 Limitations and Sources of Error

6.3 Future Research Directions

Appendix A

An example of an appendix

This is what an appendix looks like!

Appendix B

Session Protocol

Zigzag Session Protocol

Samuel Meijer

March 2019

1 Introduction and Motivation

In order to understand how to steer a boat, it is necessary to understand how the boat responds to rudder inputs. The methods used in this protocol to measure the response of the boat are adapted from the methods presented in the work of Abkowitz¹ and Lewis².

2 Description

The goal of this session is to perform the zigzag maneuver (also known as *Kempf Overshoot* or “Z” maneuver³) to derive a relationship between the rudder angle, δ , and the rate of rotation of the boat, $\dot{\theta}$. This involves steering the boat in a controlled zigzag pattern.

3 Procedure

The maneuver is conducted as follows:
while rowing,

1. the boat is set travelling straight (rudder angle, $\delta = 0$)
2. the rudder is turned to a set angle ($\delta = \delta_1$) for a set period of time, τ (until the change of rate of rotation, $\dot{\theta}$, is equal to zero [$\dot{\theta} = 0$])
3. the rudder is turned to a set angle in the opposite direction ($\delta = -\delta_1$)

Steps 2 and 3 are repeated a number of times the same rudder angles, δ_1 and the same length of time for the rate of rotation to approach zero, τ .

For this session, the rudder position will be controlled **automatically** by a servo running a programmed course. In this case, $\tau = 15s$ and $\delta_1 = -\delta_2 = 20^\circ$.

¹Martin A. Abkowitz. *Measurement of Ship Hydrodynamic Coefficients in Maneuvering From Simple Trials During Regular Operations*. M.I.T. Department of Ocean Engineering, Cambridge, November 1984.

²Edward V. Lewis. *Principles of naval architecture*. Society of Naval Architects and Marine Engineers, Jersey City, 2nd revision (3rd ed.) 1988.

³Ibid.

Appendix C

Software

```
1 #include <Wire.h>
2 #include <Adafruit_Sensor.h>
3 #include <Adafruit_BNO055.h>
4 #include <utility/imumaths.h>
5 #include <Adafruit_GPS.h>
6 #include <SPI.h>
7 #include <SD.h>
8 #include <Servo.h>
9 #include <math.h>
10 #include "Filter.h"
11
12 /* This driver reads raw data from the BNO055, Potentiometer and
    GPS
13
14 Connections for ADALOGGER
15 =====
16
```

```

17 Potentiometer
18 _____
19 Connect one side to VCC
20 Connect other to common ground
21 Connect Middle to A0
22
23 GPS
24 _____
25 Connect VIN to VCC
26 Connect GROUND to common ground
27 Connect GPS TX to RX1 (D0)
28 Connect GPS RX to TX1 (D1)
29
30 */
31
32 /*****/
33 /*BNO055 (orientation) setup, variables and def'n*/
34 /*****/
35 // Sample delay
36 const float SAMPLERATE = 10;
37 // Setup for the differentiation variables
38 double psierror = 0;
39 double dt = 0;
40 double dpserror_dt = 0;
41 double previouspsierror = 0;

```

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